

SHUBHAM R CHOWTA

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EDUCATION

Sahyadri College of Engineering & Management

2023 – 2027

B.Tech in Computer Science & Engineering (AI & ML)

TECHNICAL SKILLS

Languages: C, Python, Java

Core CS: Operating Systems, DBMS, Computer Networks

Systems / Backend: Syscalls, Virtual File Systems, REST APIs, Flask, Linux, WebAssembly (basic)

Machine Learning & AI: PyTorch, TensorFlow, CNNs, TensorFlow Lite (TFLite), RAG (LlamaIndex), Vector Databases

PROJECTS

Opportunistic Distributed Deep Learning on Personal Devices (Major Project — In Progress)

Python, PyTorch, FastAPI

- Designing an asynchronous distributed training framework using a centralized controller and heterogeneous, unreliable workers
- Decomposed training into time-bounded gradient computation tasks to leverage idle personal devices such as laptops and smartphones
- Implementing staleness-aware gradient aggregation and adaptive batch sizing based on device throughput
- Targeting wall-clock training speedup over single-device baselines under real-world execution constraints

RuneOS — Browser-Based Experimental Operating System (Version 1)

C, WebAssembly, JavaScript

- Implemented a monolithic kernel with syscall-style interfaces, process table, and cooperative round-robin scheduler
- Built a tree-based virtual file system with directories, files, and pseudo filesystems such as `/proc` and `/kernel`
- Developed a shell and browser-based desktop UI layered strictly on kernel APIs

Arecanut Disease Classification & Edge ML and UAV Deployment

Python, TensorFlow, TFLite, React Native

- Built and trained a CNN for arecanut disease classification using real-world agricultural imagery
- Evaluated domain transfer on UAV-captured images, observing ~90% effective accuracy under distribution shift
- Converted the TensorFlow model to TensorFlow Lite (TFLite) and deployed inference via UAV pipeline and React Native app

Agentic AI Assistant with Execution and RAG

Python, LlamaIndex

- Designed a tool-using AI agent capable of writing and executing code for mathematical and technical tasks
- Implemented scoped command execution, database operations, and persistent note storage
- Integrated retrieval-augmented generation (RAG) using vector embeddings to ground responses in domain-specific knowledge